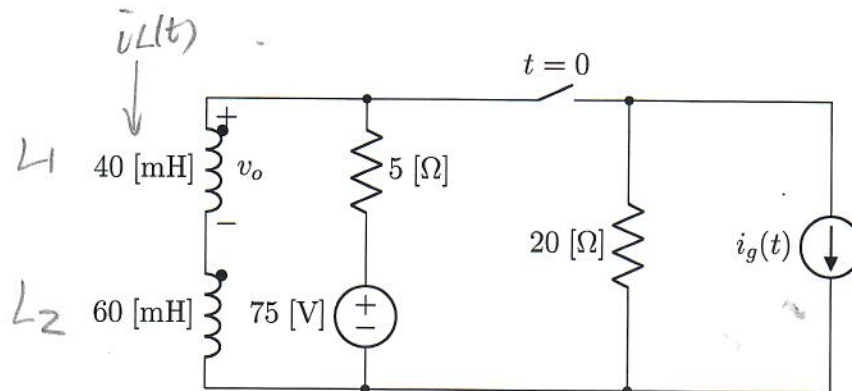


{50 Points} Consider the circuit shown below. The 10 [A] current source is replaced with a variable source $i_g(t)$. The two inductors (40 [mH] and 60 [mH]) are placed in close proximity, creating a mutual inductance M . The switch has been open for a long time and closes at $t = 0$.



Define the inductor-branch current $i_L(t)$ as flowing *downward* through the 40 [mH] and 60 [mH] pair.

Part 1 — DC Baseline (Fundamentals). Source: $i_g(t) = 10$ [A] (DC). Coupling: $M = 0$.

- 1) (5 pts) Calculate the initial energy stored in the 40 [mH] inductor at $t = 0^-$.
- 2) (5 pts) Find the final steady-state voltage $v_o(\infty)$ across the 40 [mH] inductor.

$$1) \quad \bar{v}_L(t) \Big|_{t=0^-} = \frac{75V}{5\Omega} = 15 A$$

Initial energy stored.

$$\begin{aligned} W_{(40mH)} &= \frac{1}{2} L_1 \dot{v}_L(t)^2 \\ &= \frac{1}{2} \cdot 40 \times 10^{-3} H \cdot (15A)^2 \\ &= 4.5 J \end{aligned}$$

2) In DC state, $t \rightarrow \infty$,

$$\frac{d\dot{v}_L(t)}{dt} \Big|_{t \rightarrow \infty} = 0$$

$$\begin{aligned} \therefore v_o(\infty) &= L \cdot \frac{d\dot{v}_L(t)}{dt} \\ &= 40 \times 10^{-3} \cdot 0 \\ &= 0 \quad [V] \end{aligned}$$

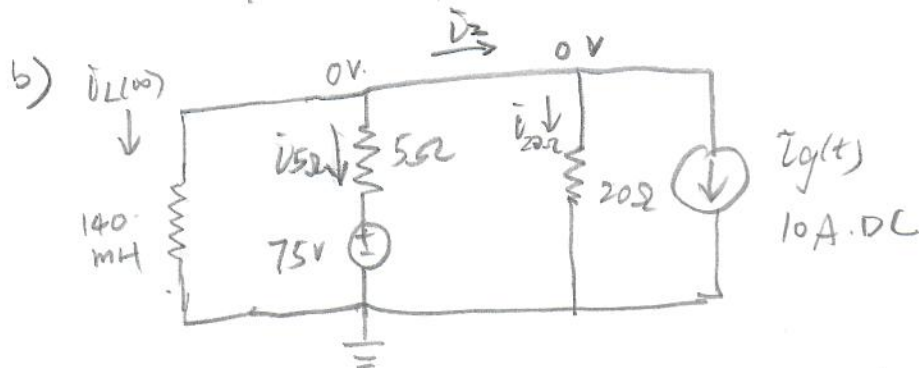
Part 2 — The Coupled System (Transients). Source: $i_g(t) = 10$ [A] (DC). Coupling: $M = 20$ [mH].

1) (5 pts) Calculate the new equivalent inductance (L_{eq}) of the series pair.

2) (5 pts) Determine the new total energy stored in the system as $t \rightarrow \infty$.

1) L_1 and L_2 , check the dot convention, the current goes in from dotted sign.

$$\begin{aligned} L_{eq} &= (L_1 + M) + (L_2 + M) \\ &= 40 \text{ mH} + 20 \text{ mH} + 60 \text{ mH} + 20 \text{ mH} \\ &= 140 \text{ mH} \end{aligned}$$



At DC state, the voltage drop on the inductors are zero.

So the current on the 20Ω Resistor is zero:

$$i_{20\Omega} = \frac{0-0V}{20\Omega} = 0 \text{ A}$$

the current on the 5Ω : $i_{5\Omega} = \frac{0-75V}{5\Omega} = -15 \text{ A}$

$$i_2 = i_{20\Omega} + i_g = 0 + 10 \text{ A} = 10 \text{ A}$$

Using KCL on the node on top of the 5Ω :

$$i_L(t) + i_{5\Omega} - i_2 = 0$$

$$i_L(t) = i_2 + i_{5\Omega} = 10 \text{ A} - 15 \text{ A} = -5 \text{ A}$$

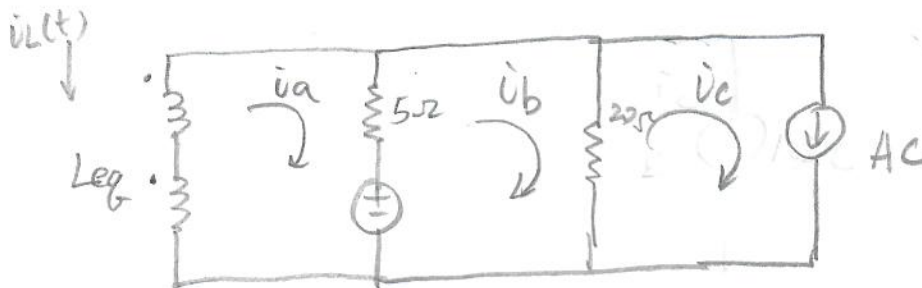
So, the energy stored in system:

$$W = \frac{1}{2} L_{eq} \cdot i_L^2 = \frac{1}{2} \cdot 140 \text{ mH} \cdot (-5 \text{ A})^2 = 1.75 \text{ J}$$

(if you define
your reference
 i_L up,
you will get
+5A)

Part 3 — Mesh-Current Analysis & Mixed-Signal Steady State (AC/DC). Source: $i_g(t) = 10 \cos(1000t)$ [A]. Coupling: $M = 20$ [mH].

- 1) (5 pts) Define mesh currents for the circuit. Label them clearly on a redrawn diagram and specify their reference directions.
- 2) (10 pts) Write a complete set of equations for this circuit using the mesh-current method.
- 3) (15 pts) Find the steady-state expression for the current $i_L(t)$.
Hint: Use the Principle of Superposition to combine the effects of the DC voltage source and the AC current source.



Left branch mesh:

$$Leq \frac{di_L}{dt} + 5(i_a - i_b) - 75 = 0$$

mesh b:

$$+75 + 5(i_b - i_a) + 20(i_b - i_c) = 0$$

mesh c:

$$20(i_c - i_b) + 0 = 0$$

$$\text{and } i_c = i_g(t)$$

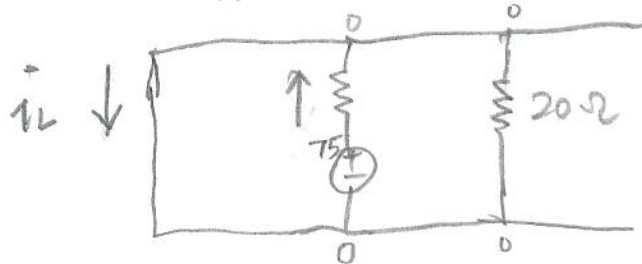
Extra space for Problem 1.

DC part: contribution

3) zero out AC, open circuit of AC source.

At DC steady state, inductors are short.

$$\bar{I}_L, DC = 15 A$$

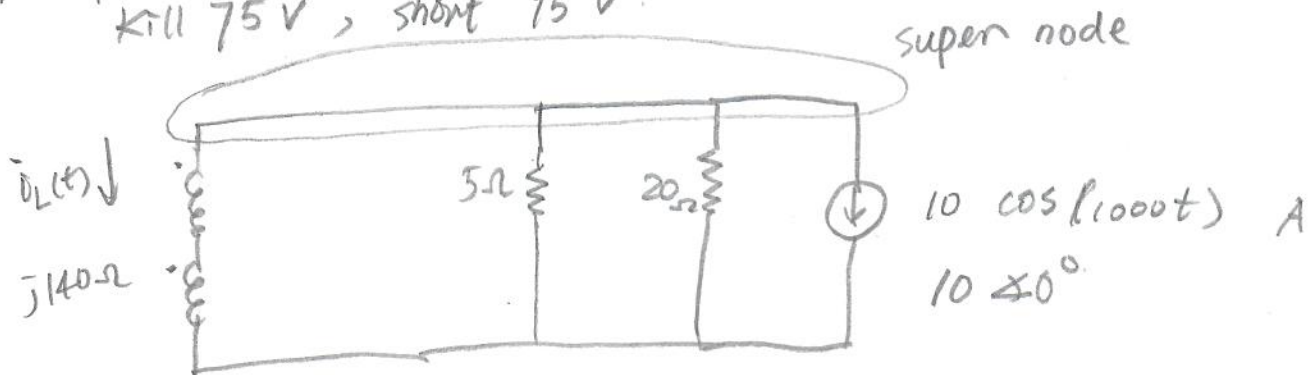


No current in the 20Ω branch. only 15A from the 5Ω branch.

$$\text{So } \bar{I}_L, DC = 15 A$$

AC part: contribution.

Kill 75V, short 75V.



$$\text{at } \omega = 1000, j\omega L_{eq} = j \cdot 1000 \cdot 140 \text{ mH} = j 140 \Omega$$

Now solve it in phasor.

Let V be the phasor voltage. (top to bottom)

$$\frac{V}{j140} + \frac{V}{5} + \frac{V}{20} + 10 = 0$$

at the super node on the top.

$$\text{Solve } V: V \left(\frac{1}{j140} + \frac{1}{5} + \frac{1}{20} \right) = -10$$

$$\therefore V = 40 \angle 181.6^\circ [V] \text{ Peak value. } 40 V$$

$$\therefore \bar{I}_{L, AC} = \frac{V}{j140} = 0.286 \angle 91.6^\circ A$$

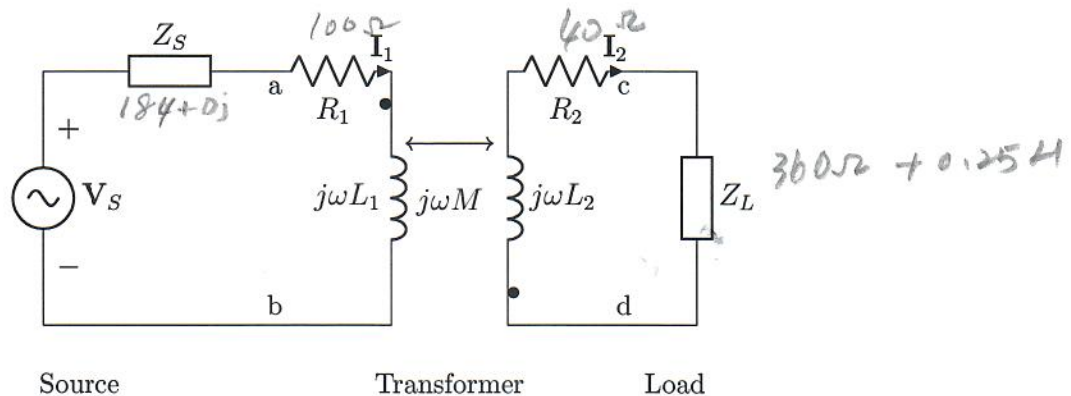
Next, change to time domain

$$\bar{v}_L(t)_{AC} = 0.286 (1000t + 91.6^\circ) \quad (A)$$

Finally, combine the DC and AC components.

$$\bar{v}_L(t) = 15 + 0.286 \cos(1000t + 91.6^\circ) \quad A.$$

{50 Points} A linear transformer couples a load consisting of a $360\ \Omega$ resistor in series with a $0.25\ \text{H}$ inductor to a sinusoidal voltage source, as shown in the figure below. The voltage source has an internal impedance of $Z_S = (184 + j0)\ \Omega$ and a maximum voltage of $245.2\ \text{V}$, and it is operating at $\omega = 800\ \text{rad/s}$. The transformer parameters are $R_1 = 100\ \Omega$, $L_1 = 0.5\ \text{H}$, $R_2 = 40\ \Omega$, $L_2 = 0.125\ \text{H}$, and $k = 0.4$.



- (10 pts) Calculate the reflected impedance Z_r seen by the primary side of the transformer.
- (5 pts) Calculate the primary current $\mathbf{I}_1$ (rms phasor).
- (5 pts) Calculate the secondary current $\mathbf{I}_2$ (rms phasor).
- (10 pts) Find the Thévenin equivalent with respect to the terminals c, d .
- (10 pts) Calculate the average power delivered by the source and the average power dissipated by the load.
- (5 pts) Calculate the power efficiency η .
Hint: efficiency is the ratio of useful power delivered to the load versus the total power supplied by the source.
- (5 pts) If the coupling coefficient is doubled to $k = 0.8$, what is the new power efficiency η' ?

Room for extra work

Extra space for Problem 2.

Given $Z_s = 180 + j0 \Omega$

$V_{s \max} = 245.2 \text{ V}$, so $V_{s, \text{rms}} = \frac{245.2}{\sqrt{2}} = 173.38 \text{ V}$

$\omega = 800 \text{ rad/s}$.

$R_1 = 100 \Omega$, $L_1 = 0.5 \text{ H} \Rightarrow j\omega L_1 = j400 \Omega$

$R_2 = 40 \Omega$, $L_2 = 0.125 \text{ H} \Rightarrow j\omega L_2 = j100 \Omega$

$k = 0.4 \Rightarrow M = k\sqrt{L_1 L_2} = 0.4 \cdot \sqrt{0.5 \cdot 0.125} = 0.4 \cdot \sqrt{\frac{1}{2} \cdot \frac{1}{8}} = 0.1 \text{ H}$.

$\therefore j\omega M = j \cdot 800 \cdot 0.1 \text{ H} = j80 \Omega$.

Load: $Z_L = 360 + j\omega \cdot 0.25 = 360 + j200 \Omega$.

Dot convention:

I_1 enter dot

I_2 enter dot

so additive coupling

a) $Z_r = \frac{(\omega M)^2}{(R_2 + j\omega L_2) + Z_L} = \frac{(800 \cdot 0.1)^2}{(40 + j100) + (360 + j200)}$

$= \frac{80^2}{500 \angle 36.87^\circ} = 12.8 \angle -36.87^\circ$
 $= 10.24 - j7.68 \Omega$

b) Total impedance in primary branch.

$Z_{\text{pri}} = Z_s + (R_1 + j\omega L_1) + Z_r$

$= (180 + j0) + (100 + j400) + (10.24 - j7.68)$

$= 290.24 + j392.32$

$= 490 \angle 53.13^\circ$

$\therefore I_1 = \frac{V_{s, \text{rms}}}{Z_{\text{pri}}} = \frac{173.38 \text{ V}}{490 \angle 53.13^\circ} = 0.354 \angle -53.13^\circ \text{ A}_{\text{rms}}$

Extra space for Problem 2.

c). check the dot, the voltage drop from the same dot sign.

$j\omega M \cdot I_1$ voltage drop from primary's mutual effect



$$I_2 = - \frac{j\omega M I_1}{R_2 + j\omega L_2 + Z_L}$$

$$= - \frac{j \cdot 800 \cdot 0.1 \cdot I_1}{400 + j100 + 360 + j200}$$

$$= -j \frac{80}{400 + j300} \cdot 0.357 \angle -53.13^\circ$$

$$= 0.057 \angle -180^\circ \text{ A (rms)}$$

$$= -0.057 \text{ Arms}$$

Extra space for Problem 2.

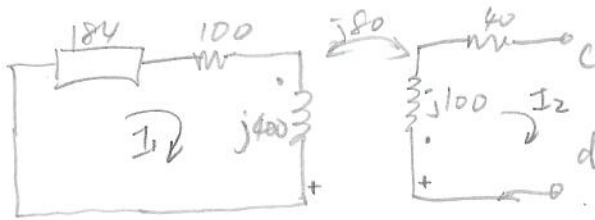
d) Calculate Thevenin equivalent at c, d

Step 1 Z_{th} : Kill V_s

The primary self-impedance $Z_{11} = Z_s + R_1 + j\omega L_1 = 284 + j400$

The impedance reflected from primary to secondary,

$$Z_r' = \frac{(\omega M)^2}{Z_{11}} = \frac{(80)^2}{284 + j400} = 7.55 - j10.64 \Omega$$



Looking into terminal c, d (remove load)
the impedance.

$$Z_{th} = R_2 + j\omega L_2 + Z_r'$$

$$= 40 + j100 + (7.55 - j10.64)$$

$$= 47.55 + j89.36 \Omega$$

$$= 101 \angle 62.7^\circ$$

Step 2. V_{th}

With c, d open, $I_2 = 0$

No mutual effect from secondary to primary.

$$I_1 \Big|_{\text{open circuit}} = \frac{V_s}{Z_{11}} = \frac{173.38 \angle 0^\circ \text{ (rms)}}{284 + j400} = 0.353 \angle -54.64^\circ \text{ (rms)}$$

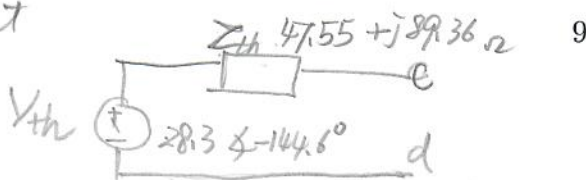
The open-circuit voltage at c-d is just the induced voltage across L_2 (secondary) by I_1

Note, voltage drop at dot ^(top) sign on primary, also drop at dot (bottom) on secondary. $V_c - V_d = -j\omega M I_1$

$$I_2 = -j\omega M \cdot I_1 \Big|_{\text{open-circuit}}$$

$$= (80 \angle -90^\circ) \cdot (0.353 \angle -54.64^\circ) = 28.3 \angle -144.6^\circ \text{ (Vrms)}$$

Equivalent



e) Average power.

$$P_{\text{source}} = \operatorname{Re}\{V_s I_1^*\}, \text{ use Rms value.}$$

$$= \operatorname{Re}\left\{ \frac{245.2}{\sqrt{2}} \angle 0^\circ \cdot 0.354 \angle 53.13^\circ \right\}$$

$$= 36.8 \text{ W}$$

$$P_{\text{load}} = |I_2|^2 \cdot R_{\text{load}} \quad (\text{use Rms value}), \text{ only } R \text{ absorbs}$$

$$= (0.0566)^2 \cdot 360 \Omega$$

$$= 1.15 \text{ W}$$

f) Power efficiency.

$$\eta = \frac{P_{\text{load}}}{P_{\text{source}}} = \frac{1.15 \text{ W}}{36.8 \text{ W}} = 3.13 \%$$

g) Now $k = 0.8$

$$M' = k \sqrt{L_1 L_2} = 0.8 \sqrt{(6.5)(0.125)} = 0.2 \text{ H}$$

$$j\omega M' = j800 \cdot 0.2 = j160 \Omega$$

Z_{22} : unchanged.

$$Z_r' = \frac{(\omega M')^2}{Z_{22}} = \frac{(160)^2}{400 + j300} = 40.96 - j30.72 \Omega$$

$$Z_{\text{primary}}' = Z_s + R_1 + j\omega L_1 + Z_r' = 324.96 + j369.28 \Omega = 491.9 \angle 48.65^\circ$$

$$I_1' = \frac{V}{Z_{\text{primary}}'} = \frac{173.38}{324.96 - j369.28} = 0.2329 - j0.2646 \text{ A} = 0.353 \angle -48.68^\circ$$

$$I_2' = \frac{(j\omega M) I_1'}{Z_{22}}$$

$$P_{\text{load}}' = |I_2'|^2 \cdot 360 = 4.58 \text{ W} \quad P_{\text{source}} = \operatorname{Re}\{V \cdot I_1'^*\} = |V| \cdot |I_1'| \cdot \cos(0 + 48.68^\circ) = 40.37 \text{ W}$$

$$\eta' = \frac{P_{\text{load}}}{P_{\text{source}}} = \frac{4.58}{40.37} = 11.3 \%$$